

Hang Yu

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Overview

My current research focuses on **Human-Centered AI** in **Robotics**, specifically, on learning from multi-modular teachings, including human feedback (**RLHF**), demonstrations (**LfD**), and generative methods(**VLM**, **VLA**). He is now a research intern at ABB Robotics, working on VLAs for long-horizon tasks and **Agentic Robotics**.

Skills

Robotic Skills: ROS, ROS2, Gazebo, Isaac Sim, Gello, UMI, VLA, Sim2Real

Robotic Hardware: Kinova Gen2/Gen3/Gen3-lite, ABB IRB 1300, Fetch, UR3E, UR5, Misty II, Cozmo

Machine Learning Skills: RL, Inverse RL, RLHF, LfD, RecSys, Imitation Learning, LLM/VLM/VLA Training

Education

Tufts University

Ph.D. in Computer Science, GPA: 4.0/4.0

Advisor: Dr. Elaine Schaertl Short

Medford, MA

Dec. 2026(expected)

Tufts University

M.S.E. in Computer Science, GPA: 3.95/4.0

Advisor: Dr. Elaine Schaertl Short

Medford, MA

Jan. 2021

Research/Work Experience

Research Intern, ABB Robotics

Build data effective VLAs for bimanual manipulation for long-horizon tasks.

Exploring agentic robotics for industrial robot applications.

June 2026 – Present

Research Scientist Intern, Berkshire Grey, SoftBank Group

Work on robot learning and developing VLA for bimanual-manipulation robotic arms.

Build Gello and UMI pipeline for collecting data for VLA training and fine-tuning.

Jan 2026 – June 2026

Research Assistant, Assistive Agent Behavior and Learning Lab, Tufts University

Conduct research on robot learning and human-in-the-loop learning.

Robot learning from non-expert users with data collected in real-world scenarios in public spaces.

2020 – present

Publications

Peer-Reviewed Publications

- C6 **Hang Yu**, James Staley, Shijie Fang, Jindan Huang, Wenchang Gao, Reuben M. Aronson, and E. Short, *PHIRL: Aligning Rewards with Progress for Inverse Reinforcement Learning with Vision-Language-Model-in-the-Loop*, (**CoRL**), 2026 (under review).
- DC1 **Hang Yu**, Elaine Short, *Enabling Robust Learning from Non-Experts by Leveraging Human Demonstrations and Human Feedback*. IEEE International Conference on Robotics Automation **ICRA**, Doctoral Consortium, 2025, Atlanta, USA
- C5 **Hang Yu**, Reuben M. Aronson, Katherine H. Allen, and E. Short, *From “Thumbs Up” to “10 out of 10”: Reconsidering Scalar Feedback in Interactive Reinforcement Learning* 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), Detroit, USA, 2023.
- C4 **Hang Yu**, Qidi Fang, Shijie Fang, Reuben M. Aronson, and E. Short, *How Much Progress Did I Make? An Unexplored Human Feedback Signal for Teaching Robots* 2024 IEEE **RO-MAN**, Pasadena, USA
- C3 Matthew Ebisu*, **Hang Yu***, Reuben M. Aronson, Elaine S. Short, *See What I Mean? Expressiveness and Clarity in Robot Display Design*, IEEE **RO-MAN 2025**, Eindhoven, the Netherlands.
- C2 Qidi Fang*, **Hang Yu***, Shijie Fang, Jindan Huang, Qiuyu Chen, Reuben M. Aronson, Elaine S. Short, *CHARM: Considering Human Attributes for Reinforcement Modeling*, IEEE **RO-MAN 2025**, Eindhoven, the Netherlands.
- C1 Shijie Fang*, **Hang Yu***, Qidi Fang, Reuben M. Aronson, Elaine Schaertl Short, *Demonstration Sidetracks: Categorizing Systematic Non-Optimality in Human Demonstrations*, IEEE **RO-MAN 2025**, Eindhoven, the Netherlands.

(*) means these authors contributed equally to the work.

Journal Publications.....

- J3 Tan, Z., **Yu, H.**, Wei, W., & Liu, J.. *Top-K interesting preference rules mining based on MaxClique*. **Expert Systems with Applications**, 143, 113043.
- J2 **YU Hang**, WEI Wei, TAN Zheng, LIU Jing-lei. *Contextual Preference Collaborative Measure Framework Based on Belief System*. **Computer Science**, 2020, 47(4): 74-84.
- J1 TAN, Z., LIU, J., & **YU, H.**. *Conditional preference mining based on MaxClique*. **Computer Applications**, 37(11), 3107.

Selected Workshop Papers.....

- W2 **Hang Yu**, James Staley, Shijie Fang, Wenchang Gao, Reuben M. Aronson, and Elaine S. Short *PHIRL: Progress Heuristic for Inverse Reinforcement Learning*. RSS workshop 2025: Continual Robot Learning from Humans
- W1 **Hang Yu** and Elaine Schaertl Short. 2021. *Active Feedback Learning with Rich Feedback*. HRI '21 Companion. Association for Computing Machinery, New York, NY, USA, 430–433.

Projects

Agentic Robotics

An agentic robotic skill for robots' self-training and improving in the loop.

Github: <https://github.com/HangYu8123/AgenticRobotics.git>

A World Model Powered VLA with Hierarchical Self-Attention for Manipulation

Integrating Hierarchical Self-Attention into GigaBrain-0 to achieve better grasping and object manipulation.

Github: https://github.com/HangYu8123/gigabrain_with_Hierarchical_Self_Attention

Harness Coding Workflow, GitHub 450+ Star ★

Agentic Harness for looping, coding, debugging, refactoring, and logging.

Github: <https://github.com/HangYu8123/HarnessFlow.git>

PHIRL: Progress-Heuristicized Inverse Reinforcement Learning

An IRL learning framework that learns from human feedback, human demonstrations, and VLMs.

Github: <https://github.com/HangYu8123/PHIRL-Progress-Heuristic-for-Inverse-Reinforcement-Learning>

Honors/Awards

ICRA Doctoral Consortium

Selected to participate in the ICRA Doctoral Consortium.

2025

RO-MAN Student Grant

Selected as the winner of the travel grant of RO-MAN 2024.

2024

Lanqiao Programming Competition National First Prize

Awarded to participants who were **Top 0.5%** among all the participants.

2016

Selected Professional Service

Mentor

Isabella Bock, Undergraduate Student, Summer Intern, now at Tufts University

Matthew Ebisu, Master's Student, Co-author of C3, now at MassRobotics

Shijie Fang, Master's Student, Co-author of C4, C1, C2, now at Dongnan University

Teaching Assistant

Ethics for AI, Robotics, and Human-Robot Interaction, Tufts University

Spring 2024, Spring 2025

Human-Robot Interaction, Tufts University

Fall 2022

Selected Service

ICRA Undergraduate Research Panelist

2025

Keynote Presenter at First Robotics Competition Conference NE

2024

DIAMOND Program Panelist

2023

Tufts Computer Science Student Council

2023 – Present

Program Committee for AAMAS

2026

Reviewer for International Conference on Robotics and Automation (ICRA)

2023, 2024, 2025